

PASCAL VAN DER VAART

EDUCATION

Leiden, Netherlands

PhD Candidate, TU Delft *sept 2021 - present (expected graduation: sept 2025)*
"Epistemic Uncertainty in Model-Free Deep Reinforcement Learning", supervised by Matthijs Spaan and Neil Yorke-Smith. *Coursework*: Stochastic Gradient Techniques in Optimization and Learning

MSc. Applied Mathematics, Leiden University (*cum laude*) *sept 2019 - nov 2021*
GPA: 8.6/10. *Coursework*: machine learning theory, asymptotic statistics, measure theoretic probability, applied deep learning, reinforcement learning, numerical linear algebra, continuous optimization, stochastic processes

BSc. Mathematics, Leiden University (*cum laude*) *sept 2016 - aug 2019*
GPA: 8.2/10. *Coursework*: mathematical statistics, numerical mathematics, measure theory, random walks

BSc. Physics, Leiden University (*cum laude*) *sept 2016 - aug 2019*
GPA: 8.3/10.

PUBLICATIONS

- [1] **Van der Vaart, Pascal R.**, Matthijs T.J. Spaan, and Neil Yorke-Smith. Epistemic bellman operators. In *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence, AAAI'25*. AAAI Press, 2025.
- [2] **Van der Vaart, Pascal R.**, Neil Yorke-Smith, and Matthijs Spaan. Bayesian ensembles for exploration in deep reinforcement learning. In *Proceedings of the 2024 International Conference on Autonomous Agents and MultiAgent Systems, AAMAS '24*, Richland, SC, 2024. International Foundation for Autonomous Agents and Multiagent Systems.
- [3] **Van der Vaart, Pascal R.**, Anuj Mahajan, and Shimon Whiteson. Model based multi-agent reinforcement learning with tensor decompositions. *2nd Workshop on Quantum Tensor Networks in Machine Learning (NeurIPS 2021)*, 2021.
- [4] Moritz A. Zanger, **Van der Vaart, Pascal R.**, Wendelin Böhmer, and Matthijs T.J. Spaan. Contextual similarity distillation: Ensemble uncertainties with a single model. *arXiv preprint arXiv:2503.11339*, 2025.
- [5] Moritz A Zanger, Max Weltevrede, Yaniv Oren, **Van der Vaart, Pascal R.**, Caroline Horsch, Wendelin Böhmer, and Matthijs TJ Spaan. Universal value-function uncertainties. *arXiv preprint arXiv:2505.21119*, 2025.
- [6] Yaniv Oren, Moritz A. Zanger, **Van der Vaart, Pascal R.**, Matthijs TJ Spaan, and Wendelin Boehmer. Value improved actor critic algorithms. In *Seventeenth European Workshop on Reinforcement Learning*, 2025.

TEACHING/SUPERVISION EXPERIENCE

Lecture on Bayesian Model-Free Reinforcement Learning, TU Delft *mar 2025*
Designed and delivered a lecture on Bayesian model-free reinforcement learning, as part of the MSc level course DSAIT4110 "Sequential Decision Making". Formulated and corrected exam questions.

Supervision of BSc graduation projects, TU Delft *apr 2025 - jun 2025*
Supervised five bachelor students for their graduation research projects about quantification of epistemic uncertainty in deep reinforcement learning for improving exploration.

Teaching assistant, Leiden University *sept 2018- june 2020*
Guided students in exercise classes and graded homework assignments as well as exams for several courses

including programming methods (Python), statistics and multivariate analysis. Supervised two groups of four students statistical modelling.

PRESENTATIONS

Epistemic Bellman Operators, AAAI'2025 (Philadelphia)

Presented my work titled "Epistemic Bellman Operators" [1], which was accepted as an oral presentation at the main technical track of AAAI'2025 (20% of accepted papers out of 23% overall acceptance rate).

Epistemic Bellman Operators, EWRL'2024 (Toulouse)

Presented my work titled "Epistemic Bellman Operators" [1] as poster at the European Workshop for Reinforcement Learning 2024.

Epistemic Bellman Operators, TU Eindhoven

Presented my work titled "Epistemic Bellman Operators" [1] at the Data and AI cluster in Eindhoven.

Bayesian Ensembles for Exploration in Deep Q-Learning, AAMAS'2024 (Auckland)

Presented my work titled "Bayesian Ensembles for Exploration in Deep Q-Learning" [2] as a poster at AAMAS'2024, as well as an oral presentation at the AAMAS workshop "Adaptive and Learning Agents".

Bayesian Deep Q-Learning via Sequential Monte Carlo, EWRL'2023 (Brussel)

Presented a preprint of "Bayesian Ensembles for Exploration in Deep Q-Learning" [2] as a poster at the European Workshop for Reinforcement Learning 2023.

Bayesian Deep Q-Learning via Sequential Monte Carlo, Leiden University

Presented a preprint of "Bayesian Ensembles for Exploration in Deep Q-Learning" [2] at the Reinforcement Learning Group (RLG) at Leiden University.

Model Based Multi-Agent Reinforcement Learning with Tensor Decompositions, QTNML'2021

Presented my work titled "Model Based Multi-Agent Reinforcement Learning with Tensor Decompositions" [3] based on my MSc thesis at the Neurips workshop Quantum Tensor Networks in Machine Learning as a pre-recorded oral presentation for an online conference during covid.

PROJECTS

PhD Dissertation: Epistemic Uncertainty in Deep Reinforcement Learning *sept 2021 - present*

Researched Bayesian model-free reinforcement learning methods as part of the Sequential Decision Making group at the computer science faculty of TU Delft. Supervised by Matthijs Spaan and Neil Yorke-Smith

MSc Thesis: Tensor Decompositions for multi-agent Reinforcement learning *jan 2021 - nov 2021*

A MSc graduation project about tensor approximation methods with applications to reinforcement learning, leading to a position paper at the Quantum Tensor Networks in Machine Learning workshop at Neurips 2021. Supervised by Anuj Mahajan (University of Oxford) and Johannes Schmidt-Hieber (Leiden University)

BSc Thesis: Statistical Methods for Quantum State Estimation, Leiden University *jan 2019 - aug 2019*

A mathematics and physics bachelor research project, designing and implementing statistical methods involving maximum likelihood estimation and Bayesian mean estimation via Monte Carlo methods in Python for a problem in physics.

SKILLS

Programming Languages/Frameworks

proficient: Python, ML frameworks: JAX, also familiar with TensorFlow and Torch
familiar: R, Javascript, C++

Languages: English (C2 Proficiency), Dutch (native)